
Historical Networks as Treasure Maps: Finding Relevant Sources with Aid from Attestation Networks

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Abstract

The largest surviving papyrus archive from ancient Egypt—the Archive of Zenon (263–229 BCE, esp. Vandorpe 2015; Pestman 1981a, 1981b)—provides a wealth of information about people living under kings Ptolemaios II (reg. 285–246 BCE) and Ptolemaios III (reg. 246–222 BCE). As an entrusted employee of Apollonios, the finance minister of Ptolemaios II, but also in his private affairs, Zenon interacted with a wide range of people with different social status and backgrounds. As a result, the archive holds enormous potential for enriching our understanding of socio-economic dynamics played out in early Ptolemaic Egypt and beyond (recent network approaches to the archive include Alvares Freire 2023; Tambs 2023, 2025, in press; Wolff in press).

Through existing databases such as Trismegistos (Depauw and Gheldof 2014), it is possible to relatively quickly gather data for modelling ancient sources as affiliation networks. For example, Trismegistos presents lists of people attested in the Zenon papyri, that can be retrieved, restructured and modelled as 2-mode networks of people attested in ancient texts through a semi-automated data collection process (as recently explained in Tambs 2025). The present talk critically discusses the potential and shortcomings of such attestation networks for historical analysis, before proposing a mixed-method approach to exploring how Zenon’s multiple identities are manifested in the documentation.

First, a 2-mode network built from 1845 texts (Fig. 1) and visualised in Gephi (Bastian et al. 2009) is used to identify a subset of the archive that—from a network analytical perspective—appears relevant for exploring such questions. The subset is found with aid from a weighted version of the k -core centrality measure (recently tested in Bennett et al. 2024). Next, the author revisits the ancient sources to model and analyse multilayered interpersonal (1-mode) networks reflecting a sample of the archive, the selection of which is guided by the 2-mode network explorations. In doing so, she demonstrates how big(ger) attestation networks can serve as treasure maps, useful for finding sources to be approached with more qualitative network analysis and questions.

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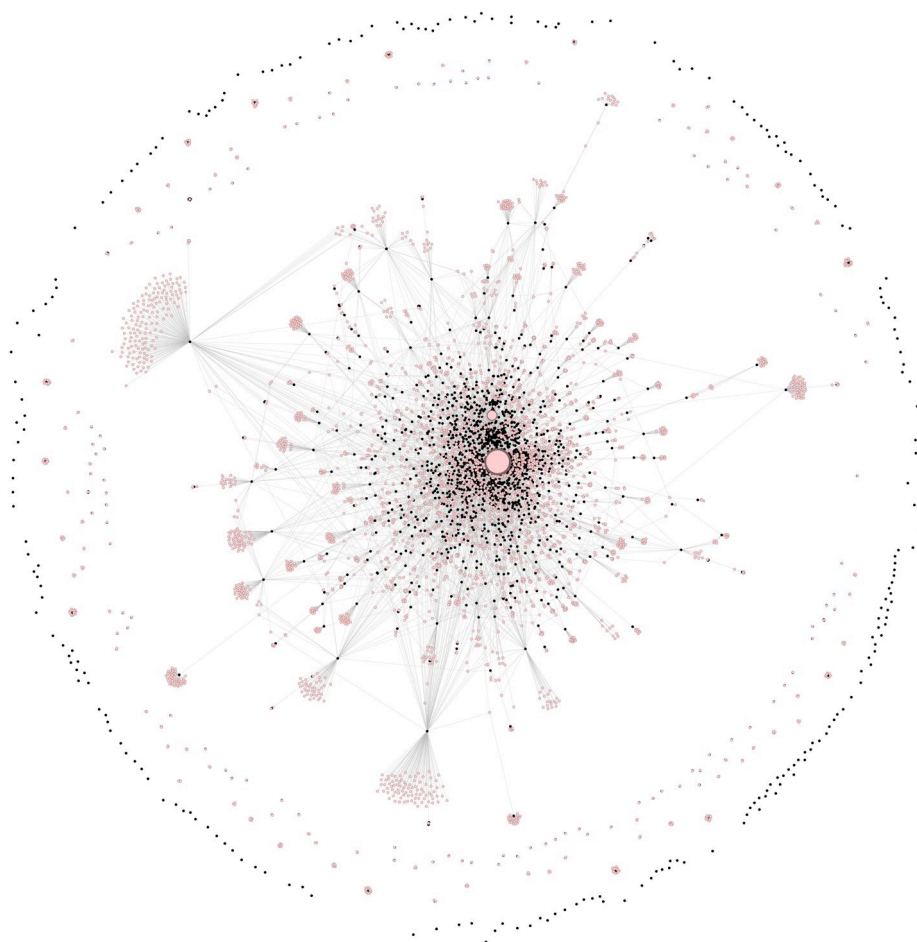


Figure 1. Directed 2-mode network showing named individuals (pink nodes) attested in (grey edges) 1845 Zenon papyri (black nodes). The model present Trismegistos data as is, collected in 2021. Graph by author, created in Gephi.